



AEROOS

RASPBERRY PI 4 / HARDWARE BRING-UP

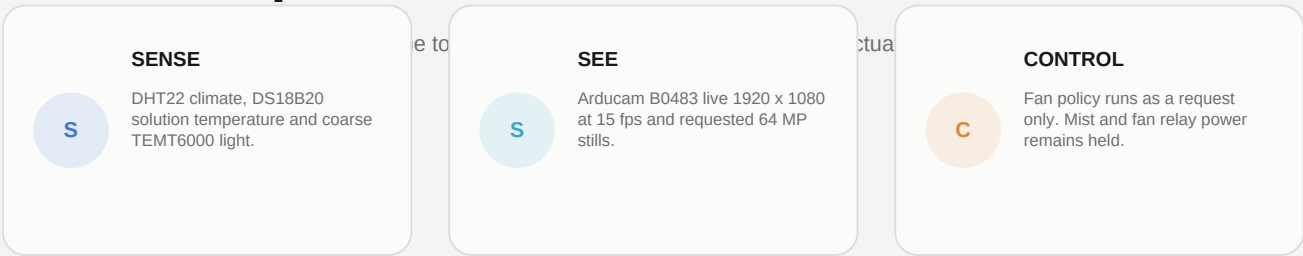
Hardware Setup and Commissioning Manual

Sensor, camera, display and dry-run control integration for the Nexora aeroponic research prototype.

REVISION 4.0 - CLOSED-LOOP CONTROL + FLASHING

Raspberry Pi 4 8 GB | SC1227 800 x 480 touch display | DHT22 | DS18B20 | TEMT6000 + PCF8591 | Arducam B0483. Includes automatic recirculation behavior, component photographs, safety gates and a visual image-flashing workflow.

What this phase does



NON-NEGOTIABLE POWER HOLD

Do not connect pumps, fans, relay load terminals, the 5 V mist maker or the future 12 V supply during this phase. Pi GPIO is 3.3 V logic only. Header 5 V pins are fixed power rails, not controllable GPIO.

Available now	Temperature, humidity, solution temperature, light percentage, camera
Visible but unavailable	pH, EC, reservoir level, delivery flow, dosing, mixer
Output condition	Actuator master enable = false; no relay GPIO object is created
Automatic recirculation	PIN-protected UI scheduler is ready; physical output stays blocked until relay commissioning
System state	Development / Interlocks unavailable

02

Identify the exact hardware first

Match each photograph to the marking on your own part. Similar-looking modules can use different pin orders and relay polarity.



Raspberry Pi 4 Model B

8 GB host; 40-pin header at the top edge.

Raspberry Pi documentation



SC1227 7 inch display

Original 800 x 480 DSI touch display.

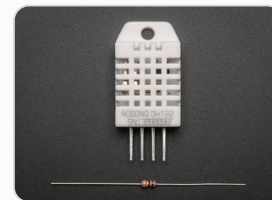
Raspberry Pi documentation



Arducam 64 MP

B0483 / IMX519 family; confirm the label on the ribbon or PCB.

Arducam 64 MP manual



DHT22 climate sensor

Use the bare-sensor or module wiring variant that matches your part.

Adafruit product 385



DS18B20 probe

Waterproof probe; wire colours are not universal.

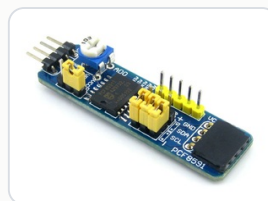
Adafruit product 381



TEMT6000 light

Three terminals: SIG, GND and VCC on the reference breakout.

SparkFun BOB-08688



PCF8591 ADC

Reference board only; verify A0, SDA, SCL, VCC and GND labels.

Waveshare PCF8591 board

PHOTO + RATING

REQUIRED

Relay and pumps

Not pictured because your models and ratings are unverified. Record every marking before use.

User hardware - photo required

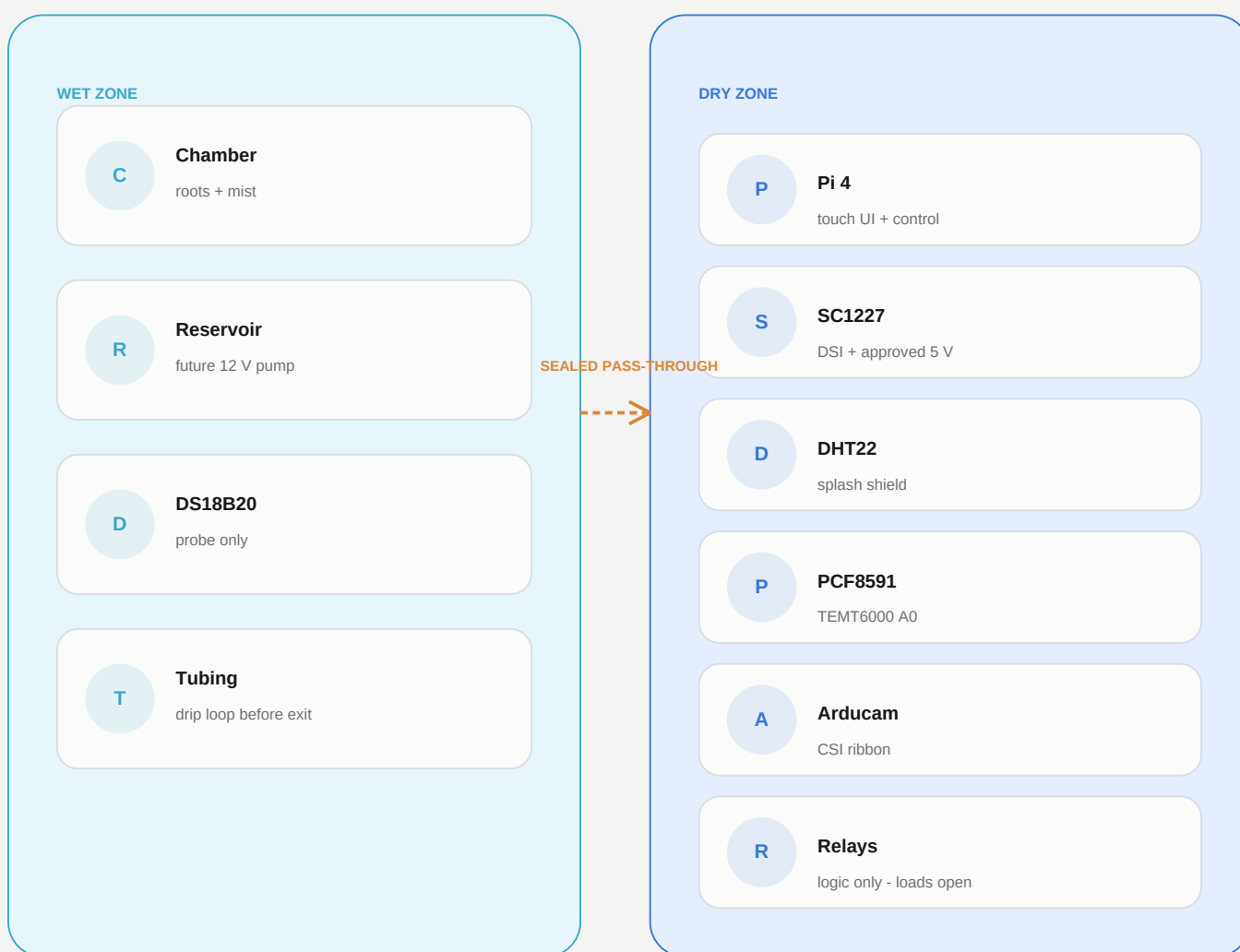
BEFORE WIRING

Photograph both sides of every relay, pump and fan. Record coil / input voltage, load voltage, rated current, terminal labels and whether the relay board claims 3.3 V logic compatibility. Stop if any marking is unreadable.

03

Separate water from logic

Treat the prototype as two physical zones joined only by protected cables and tubing.



PLACEMENT RULE

Mount the Pi, PCF8591 and relay logic higher than the reservoir rim, with connector openings facing down or sideways. Provide a drip loop on every cable approaching the dry enclosure.

04

40-pin map: physical and BCM

Orient the Pi with USB ports to the right. Pin 1 is the 3.3 V corner pin. Verify with the board silkscreen before wiring.

01	3V3	5V	02
03	GPIO2 SDA	5V	04
05	GPIO3 SCL	GND	06
07	GPIO4 1-Wire	GPIO14	08
09	GND	GPIO15	10
11	GPIO17 MIST	GPIO18 DHT22	12
13	GPIO27	GND	14
15	GPIO22	GPIO23	16
17	3V3	GPIO24	18
19	GPIO10	GND	20
21	GPIO9	GPIO25 FAN	22
23	GPIO11	GPIO8	24
25	GND	GPIO7	26
27	ID_SD	ID_SC	28
29	GPIO5	GND	30
31	GPIO6	GPIO12	32
33	GPIO13	GND	34
35	GPIO19	GPIO16	36
37	GPIO26	GPIO20	38
39	GND	GPIO21	40

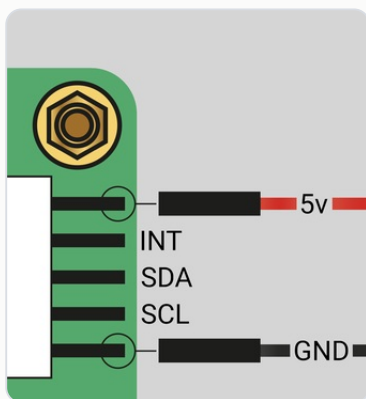
USED IN THIS PHASE

DHT22: physical 12 / BCM18. DS18B20: physical 7 / BCM4. I2C: physical 3 SDA and 5 SCL. Future relay logic: physical 11 / BCM17 and physical 22 / BCM25. Sensors use 3.3 V.

05

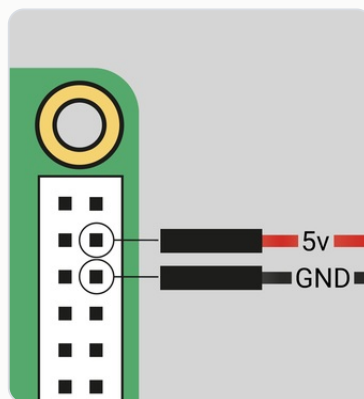
Connect the 7 inch DSI display

Use the original 15-way DSI ribbon. The display is an approved 5 V load; the pumps and relay coils are not part of this connection.



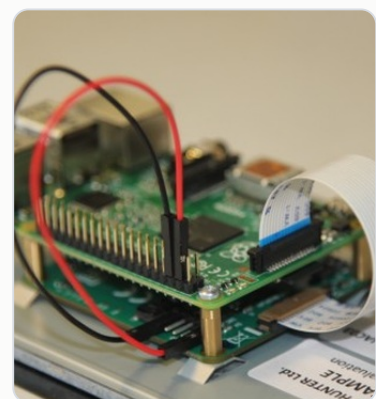
1 Display connector

Open the retaining clip evenly; insert the FFC with contacts facing up, away from the display.



2 Raspberry Pi DSI

On Raspberry Pi 4, insert the other end in the DISPLAY connector with contacts toward USB / Ethernet.



3 Display power

Red to 5 V, black to GND. Use one documented pin pair and verify twice before power-on.

Photos: Raspberry Pi documentation, Touch Display connection sequence.

ASSEMBLY SEQUENCE

- 1 Shut down; unplug USB-C power. Wait until the red power LED is off.
- 2 Fit the display-side FFC, close its latch and confirm the cable is square.
- 3 Fit the Raspberry Pi-side FFC in the connector marked DISPLAY; do not use CAMERA.
- 4 Connect display red to physical pin 2 or 4 (5 V), and black to physical pin 6 (GND).
- 5 Inspect for exposed contacts, trapped ribbon, metal contact or reversed power.
- 6 Power on and verify 800 x 480 landscape, all four touch corners and no undervoltage icon.

STOP CONDITION

If the ribbon is hot, the display flashes, touch axes are reversed or Raspberry Pi reports undervoltage: shut down, disconnect power and recheck the DSI orientation and 5 V / GND pins.

Mount and verify the 64 MP camera

Use the CSI ribbon without tight folds. Keep optics outside the direct mist plume and provide a repeatable viewing angle.



64 MP autofocus module

Use the label on your PCB or cable to confirm B0483 / IMX519 compatibility. A similar lens housing is not sufficient identification.

Photo: Arducam 64 MP camera manual.

Raspberry Pi CSI

Live: 1920 x 1080 @ 15 fps

Focus: continuous / capture cycle

Still: 9152 x 6944 on request

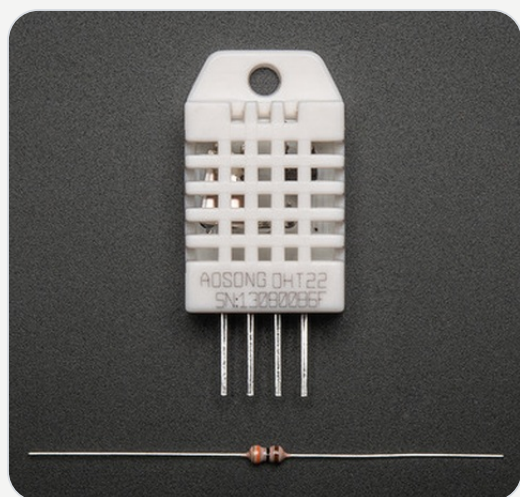
MOUNTING AND OPTICAL REFERENCE

IMAGE BUILD GATE

07

Wire the chamber climate sensor

First determine whether you have a bare four-pin DHT22 or a three-pin module. Do not copy a pin order that does not match the markings on your part.



Reference photo: Adafruit DHT22 product 385.

BARE 4-PIN SENSOR - FRONT GRILLE FACING YOU

1 VCC	3.3 V - physical pin 1 or 17
2 DATA	GPIO18 - physical pin 12
3 NC	leave open
4 GND	ground - physical pin 9
Pull-up	4.7-10 kOhm from DATA to 3.3 V

THREE-PIN MODULE

Read the PCB labels. Connect VCC to 3.3 V, OUT / DATA to GPIO18 and GND to ground. Most modules include a pull-up, but verify rather than assume.

PLACEMENT

Use a vented splash shield. Keep the sensing face at least 100 mm from nozzles and 50 mm from Raspberry Pi, buck converter or relay coils.

COMMISSIONING TEST

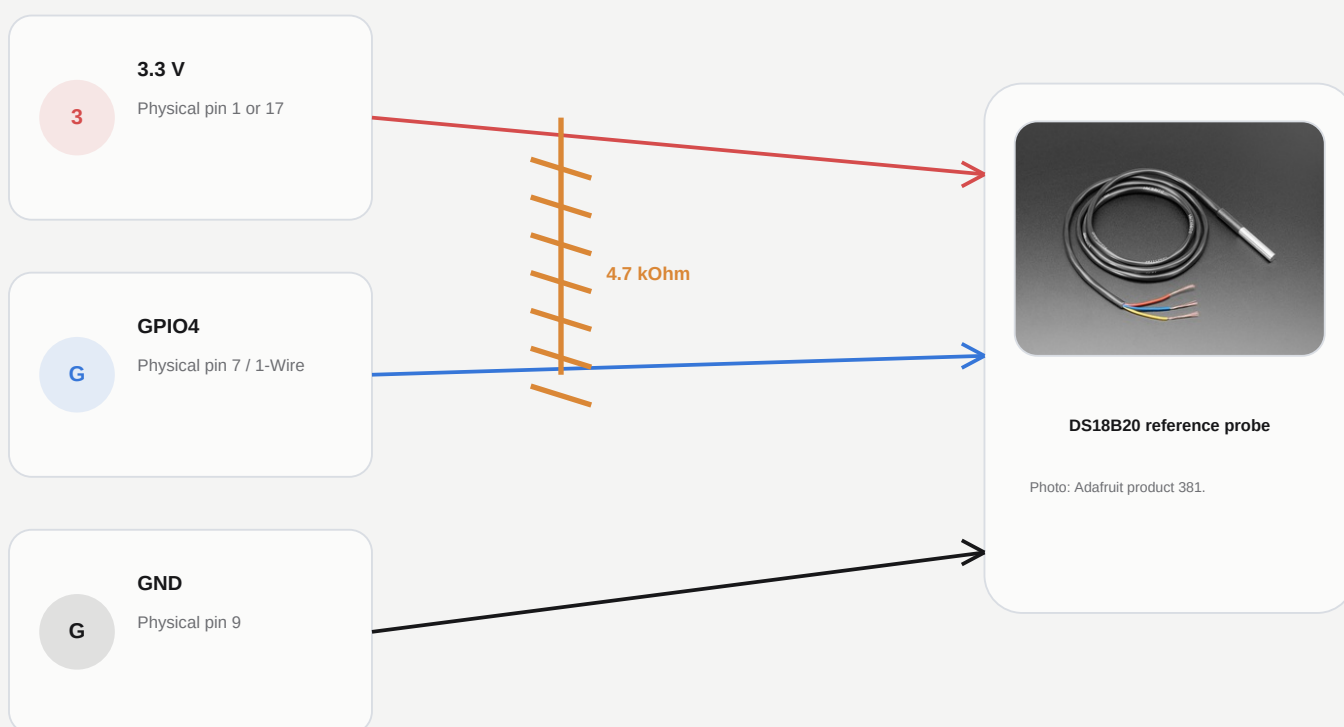
1. With power off, continuity-check VCC, DATA and GND to the intended header pins.
2. Boot; wait two minutes. Confirm readings update approximately every two seconds.
3. Compare against a trusted room reference for ten minutes; note the offset, do not silently correct it.
4. Unplug DATA: within ten seconds AeroOS must show climate unavailable and request fan OFF.
5. Reconnect: valid temperature and RH must recover without inventing the missing samples.

STOP

Any hot sensor, unstable Raspberry Pi, impossible humidity or swapped pin marking: power off and re-identify the module.

Wire the waterproof solution probe

The 1-Wire data line requires a 4.7 kOhm pull-up to 3.3 V. Probe wire colours vary; verify the seller's label before applying power.



WIRE IDENTIFICATION

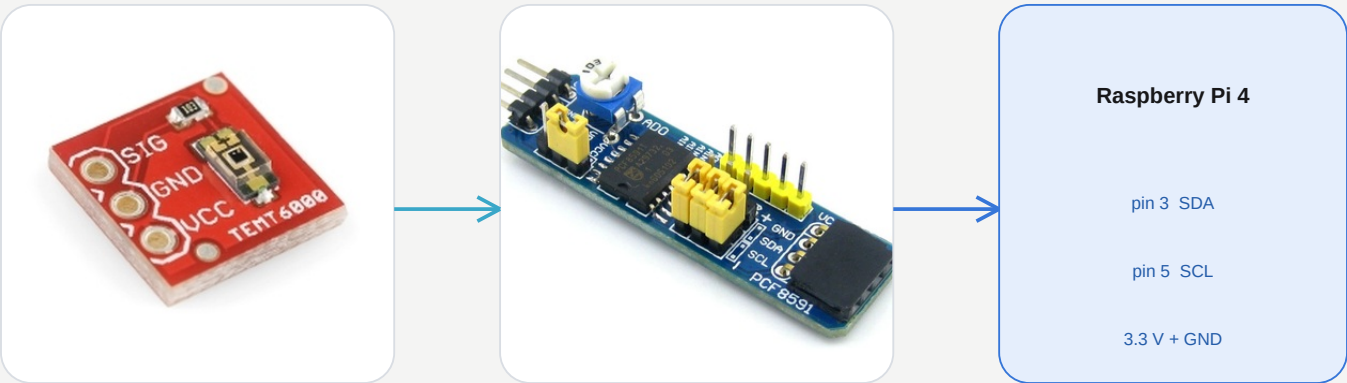
Common probes use red = VCC, black / blue = GND and yellow / white = DATA, but this is not guaranteed. Use the exact supplier diagram; never rely on colour alone.

PLACEMENT + TEST

Keep joints dry and add a drip loop. Confirm a `/sys/bus/w1/devices/28-*` device, compare at two temperatures, then unplug and verify only solution temperature becomes unavailable.

Connect coarse ambient light telemetry

The TEMT6000 analog output is read on PCF8591 A0. This 8-bit ADC is not accepted for production pH or EC chemistry.



TEMT6000 reference breakout

Photo: SparkFun BOB-08688.

PCF8591 reference board

Photo: Waveshare PCF8591 AD/DA board.

WIRING TABLE

TEMT VCC	PCF VCC	Pi 3.3 V
TEMT OUT	PCF A0	Analog channel 0
TEMT GND	PCF GND	Common Pi ground
PCF SDA	-	Physical pin 3 / BCM2
PCF SCL	-	Physical pin 5 / BCM3

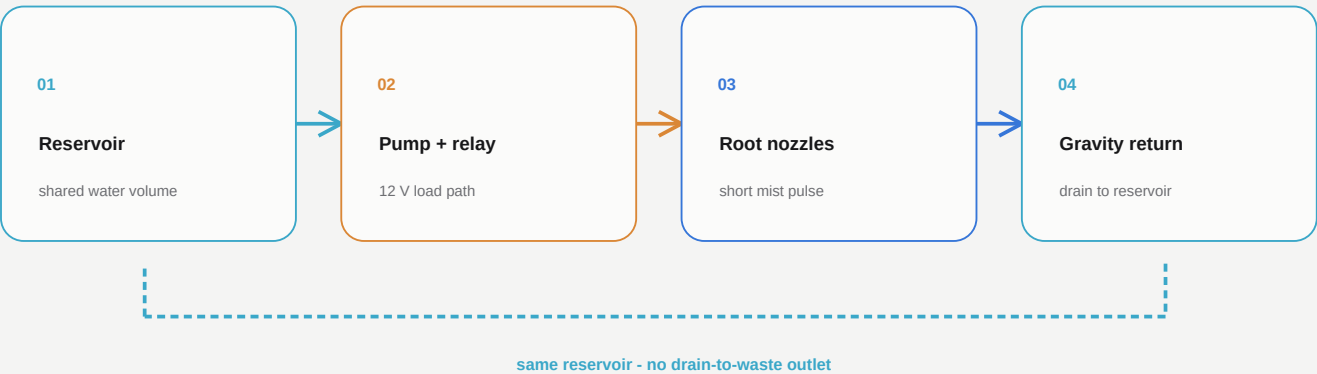
DISPLAYED VALUE

Leave A1, A2 and A3 unused. AeroOS reports a coarse 0-100% A0 percentage. Verify the labels on your own boards; a reference photo is not a pinout. The result is not calibrated lux, PAR or PPFD.

Automatic closed-loop recirculation

Water starts in one reservoir, is pumped to the root nozzles, and returns by gravity to the same reservoir. The UI switch controls the scheduler; it never directly energizes a GPIO.

PHYSICAL WATER PATH



WHAT THE INTERFACE NOW DOES

Enable switch	operator PIN required
Missing level / flow	30 min supervised session
Development pulse	maximum 2 seconds
Default interval	one request every 5 minutes
Disable switch	pump OFF immediately
Expiry / restart / fault	automation OFF + pump OFF

PHYSICAL ACTIVATION BOUNDARY

Shipped image	master enable false
Relay polarity	must be measured
Relay input	must accept 3.3 V
Pump supply	separate rated 12 V
Dry-run risk	operator supervision
Unverified state	switch remains disabled

IMPORTANT

The automatic function is implemented end-to-end, but the supplied Raspberry Pi profile deliberately keeps `actuator_master_enable=false` and both relay drivers disabled. After relay and 12 V power commissioning, the same PIN-protected switch activates the bounded scheduler. Never power a pump from a Pi header pin.

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Route cables for service and safety

Use distinct paths for low-voltage sensors, camera / display ribbons and future actuator power.

LANE A - SENSOR

DHT22, DS18B20, I2C

LANE B - HIGH SPEED

DSI display, CSI camera

LANE C - FUTURE POWER

12 V pump / fan - empty now

DRIP LOOP BEFORE ENTRY



Strain relief

Label both ends

ROUTING RULES

No ribbon cable under sharp metal edges. No sensor splice in the wet zone. Keep future motor wiring separate from I2C and 1-Wire. Leave service loops without allowing cable coils to sit below possible leaks.

12

Assembly steps 1-4

Complete each check box before moving on. Keep every power supply disconnected during continuity work.

1

Prepare and label

Photograph the Pi header. Label DHT22, DS18B20, PCF8591, TMT6000, camera, display and both relay channels.



Every lead identified at both ends

2

Mount dry enclosure

Place the Pi above the reservoir rim. Provide airflow, cable entry glands, a drip edge and access to SD / USB.



Enclosure stable and splash-separated

3

Install display

Connect DSI ribbon, approved display 5 V and ground. Keep the remaining 5 V rail clear of actuator loads.



Ribbon latched and polarity verified

4

Install camera

Connect CSI ribbon, mount outside direct mist and place a scale marker in the root field of view.



Lens clean; mount cannot shift

CONTINUITY GATE

With power still off, confirm no short between 3.3 V and GND, no short between 5 V and GND, and no accidental continuity from any relay load terminal to the Pi.

13

Assembly steps 5-8

Bring up one sensor bus at a time so a wiring fault has a narrow search area.

5

Wire DHT22

Use 3.3 V, GPIO18 and ground. Install the vented splash shield in representative chamber air.



Data line verified at physical pin 12

6

Wire DS18B20

Use GPIO4 with a 4.7 kOhm pull-up to 3.3 V. Keep the cable joint dry and add a drip loop.



Probe enumerates on 1-Wire

7

Wire light path

Connect TEMENT6000 output to PCF8591 A0, then SDA / SCL to pins 3 / 5 at 3.3 V.



0x48 appears on I2C bus 1

8

Prepare relay hold

Connect no load terminals. Confirm labels only. Leave the actuator master enable false.



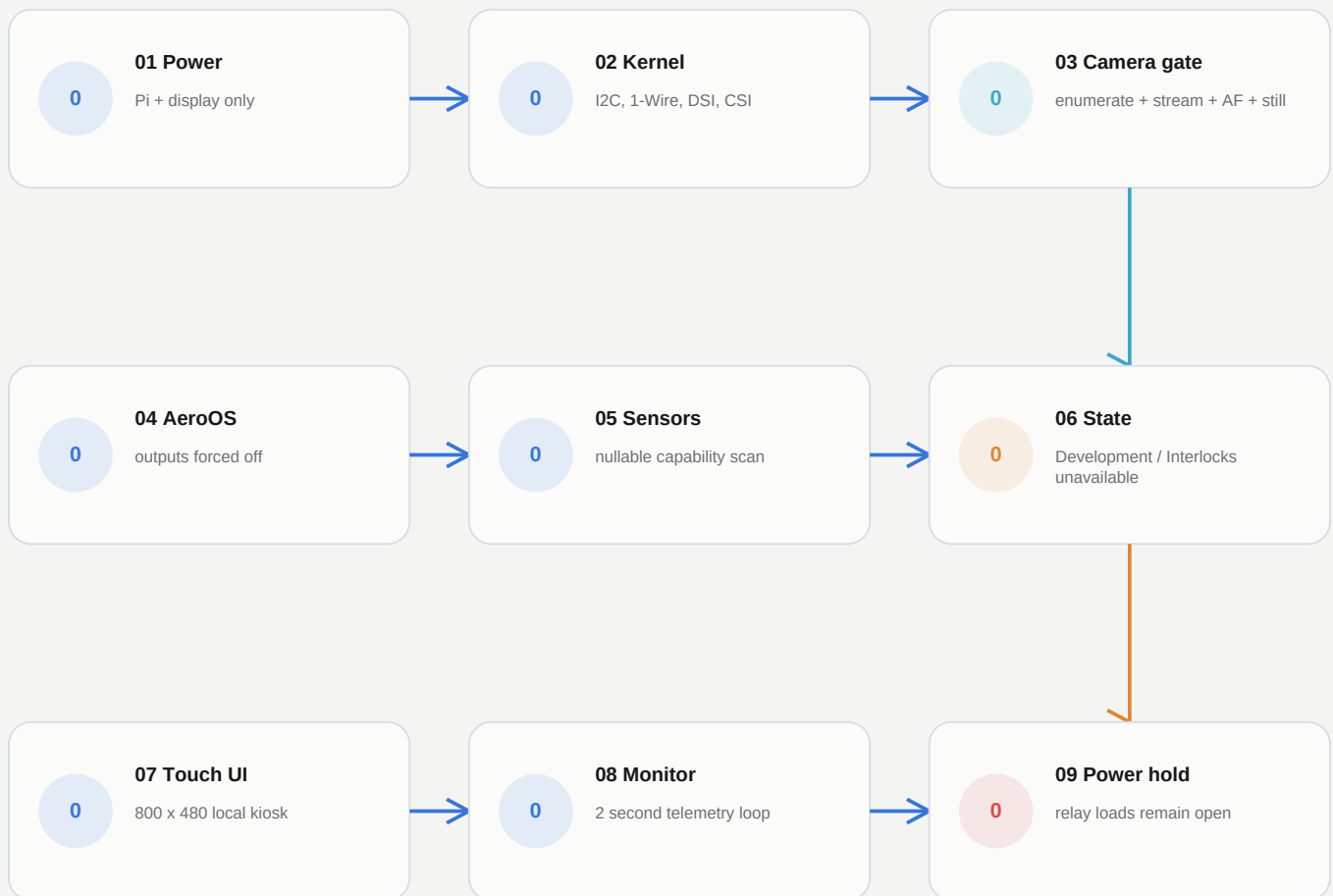
COM / NO / NC visibly marked HOLD

FIRST POWER-ON

Power the Pi and display only. Watch for heat, smell, reboot loops or undervoltage. If any appear, shut down immediately and disconnect power before inspection.

Boot and commissioning flow

The operating system proves safe state before presenting the chamber UI.



FAIL-SAFE RESULT

A camera gate failure blocks commissioning. A sensor failure is visible and nullable. Browser failure, service restart and shutdown all request outputs off.

How the automatic toggle flows

The dashboard switch controls a bounded scheduler. Physical pumping begins only after the hardware master and relay channel have been commissioned.



SHIPPED IMAGE STATE

The complete UI and scheduler can be tested in the simulator. On the Raspberry Pi image, Gate 02 is false by default, so the switch is visibly disabled and no relay GPIO object exists. Commissioning the relay and hardware master unlocks the same flow; it does not require a second UI.

Test cards

Record date, operator, result and evidence for every card. A pass requires stable behavior, not one successful sample.

DISPLAY + TOUCH

Boot 10 times. Confirm 800 x 480 landscape, accurate touch corners and no horizontal scrolling.

Evidence: Screenshot + boot log

DHT22

Observe 10 minutes. Unplug: value becomes unavailable. Reconnect: valid telemetry recovers.

Evidence: Temperature / RH history

DS18B20

Compare at room and reservoir temperature. Unplug: only solution temperature becomes unavailable.

Evidence: Two-point record

TEMT6000

Cover and illuminate sensor. Confirm monotonic 0-100% movement; never label as calibrated lux.

Evidence: ADC percentage plot

CAMERA

Stream 1080p15, autofocus, take 9152 x 6944 still, stop / restart stream and inspect sharpness.

Evidence: JPEG + gate logs

OUTPUT SAFETY

Try manual and automatic mist, fan and dosing paths. Master false must prevent every GPIO energize attempt.

Evidence: API audit + GPIO test

ACCEPTANCE

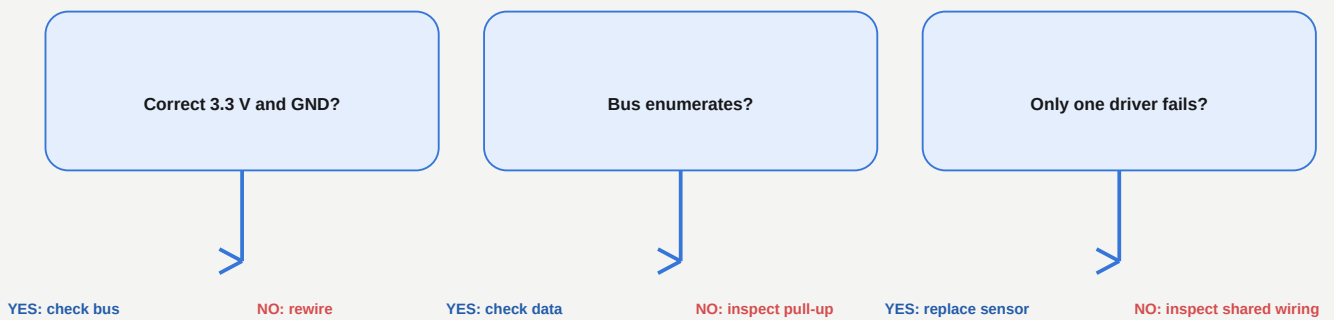
Sensors and camera operate; missing values remain unavailable; every physical actuator remains unpowered.

17

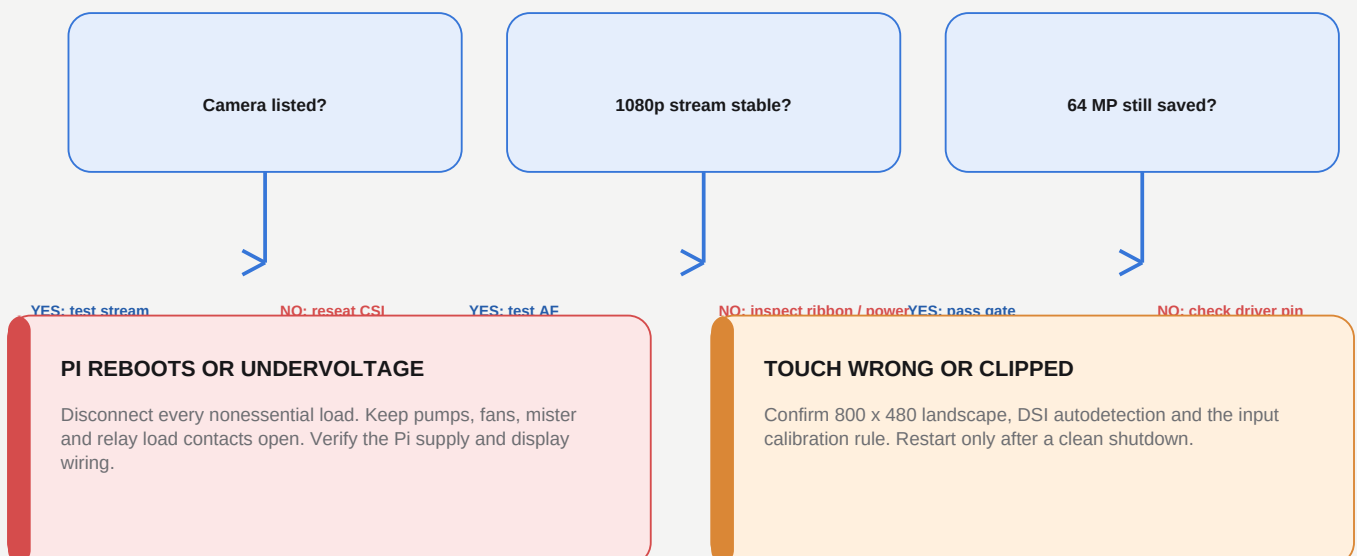
Troubleshooting decision trees

Power off before reseating any ribbon or moving any conductor.

SENSOR UNAVAILABLE

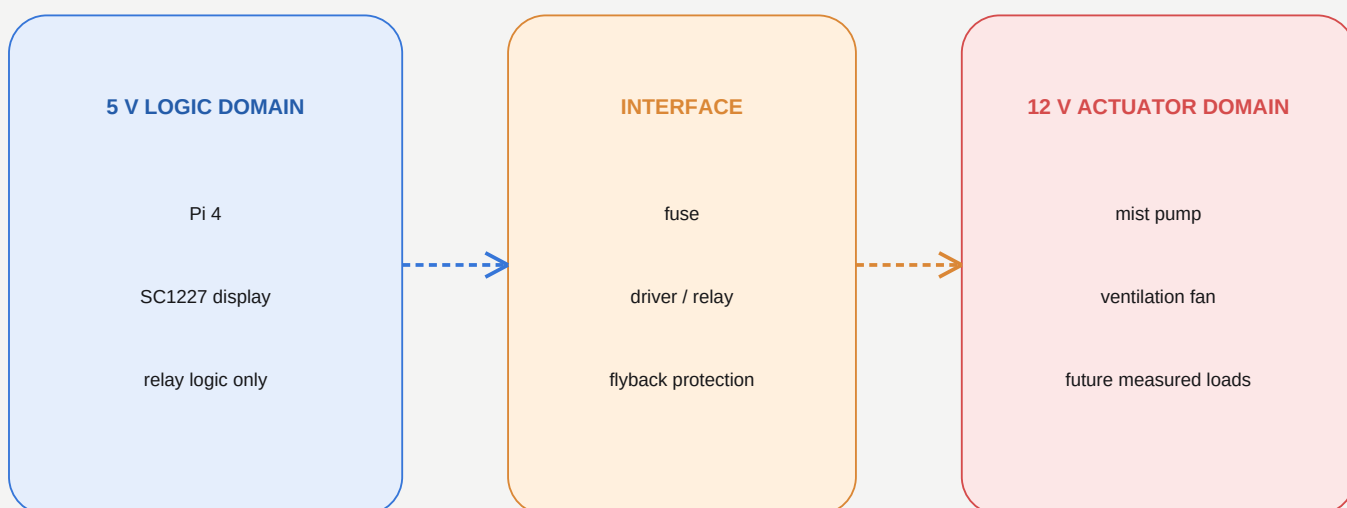


CAMERA GATE FAILED



Power architecture - future phase only

This appendix shows separation boundaries without inventing pump ratings, fuses, wire sizes or converter limits.



MEASURE BEFORE DESIGN

Record each pump and fan nameplate, steady current, startup current, voltage tolerance and duty cycle. Then select fuse, conductor, connector, driver, buck converter and enclosure thermal design.

COMMON REFERENCE

If a non-isolated logic interface requires a common ground, implement it at one documented point after verifying the module topology. Never route actuator current through Pi ground pins.

HOLD - DO NOT ASSEMBLE THIS DOMAIN IN REVISION 4.0

19

Ready for sensor and camera trials

Sign only after the evidence exists. This is not approval for powered pumps, fans, mist makers or dosing.

- ☐ Pi 4 and SC1227 boot reliably at 800 x 480 landscape
- ☐ Touch targets respond accurately across all screen edges
- ☐ DHT22 reports temperature and humidity; failure becomes unavailable
- ☐ DS18B20 reports solution temperature through 1-Wire
- ☐ PCF8591 appears at 0x48 and TEMENT6000 changes monotonically
- ☐ Arducam passes 1080p15, autofocus and 9152 x 6944 still capture
- ☐ pH, EC, level and flow remain explicitly unavailable
- ☐ Automatic recirculation is PIN-protected and master false prevents GPIO energizing
- ☐ Relay COM / NO / NC terminals are labelled HOLD - POWER PHASE
- ☐ Wet / dry separation, drip loops and strain relief are complete

PHOTO + TECHNICAL SOURCES

raspberrypi.com/documentation/accessories/display.html and [/computers/getting-started.html](https://raspberrypi.com/documentation/computers/getting-started.html); official Raspberry Pi documentation repository;
adafruit.com/product/385 and [/381](https://adafruit.com/product/381); sparkfun.com TEMENT6000 BOB-08688; waveshare.com/wiki/PCF8591_AD_DA_Board;
arducam.com/downloads/arducam_64mp_pi_camera_manual.pdf; NXP PCF8591 specification.

Operator: _____

Date: _____

Result: PASS / HOLD

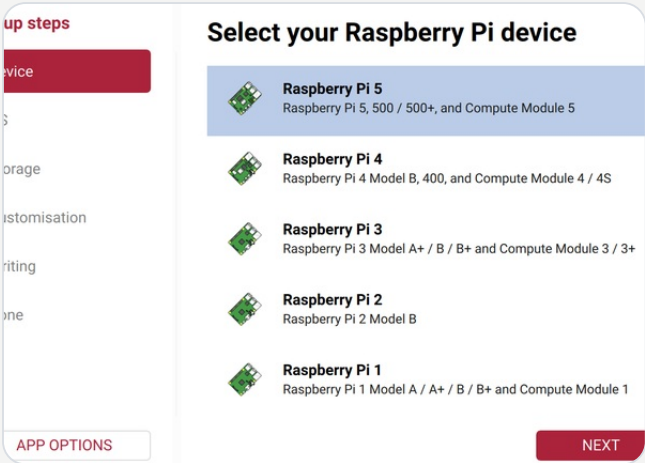
Flash the AeroOS image: select

Use Raspberry Pi Imager on macOS, Windows or Linux. The screenshots are from the official Raspberry Pi documentation; labels may move slightly between releases.

01 VERIFY AND PREPARE THE IMAGE

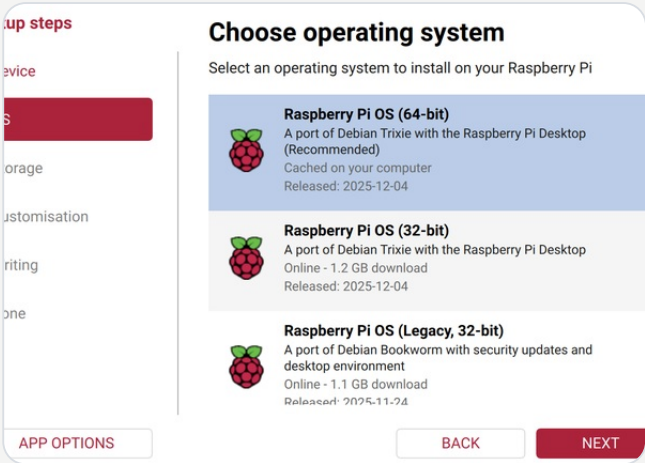
VERIFY BEFORE WRITING

In output/image, run shasum -a 256 -c SHA256SUMS-r4. If Imager does not accept the .img.zst file, create a normal .img with: unzstd -k AeroOS-RPi4-8GB-SC1227-2026-07-23-r4.img.zst. Keep at least 8 GB free for the decompressed file.



02 CHOOSE DEVICE

Select Raspberry Pi 4. Do not select Pi 5 even when it appears first.



03 CHOOSE OS > USE CUSTOM

Scroll to Use custom, then select the verified AeroOS .img. Do not choose Raspberry Pi OS.

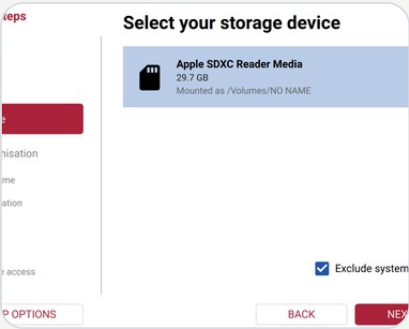
KEEP THE HARDWARE SAFE

Before flashing or first boot, disconnect pump power, fan power, relay COM / NO / NC terminals and the 5 V mist maker. The Pi, display, camera and approved 3.3 V sensors are the only connected devices for initial bring-up.

Screenshots: Raspberry Pi Ltd documentation, Getting started > Install using Imager.

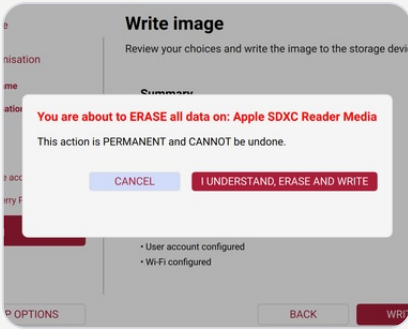
Flash the AeroOS image: write and boot

Select the microSD card by capacity, write the verified image, allow verification to finish, then perform a dry first boot.



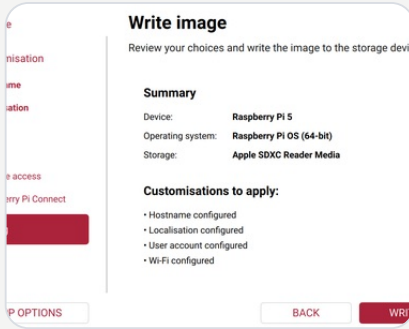
04 SELECT STORAGE

Identify the microSD by capacity. Disconnect other removable drives if uncertain.



05 CONFIRM ERASE

This destroys all data on the selected card. Recheck the device name and size.



06 WRITE + VERIFY

Do not cancel verification. Wait for the successful completion message.

DRY FIRST-BOOT SEQUENCE

1

1 Eject
Safely eject the card and insert it into the unpowered Pi.

2

2 Inspect
Confirm relay loads and 12 V pump power are still disconnected.

3

3 Power
Power the Pi + official display; use Ethernet for initial bring-up.

4

4 Gate
Wait for camera enumeration, 1080p15, autofocus and 64 MP still tests.

5

5 Dashboard
Confirm 800 x 480 touch, real sensors and unavailable missing capabilities.

6

6 Automation
The automatic switch remains disabled until relay/master commissioning.

Screenshots: Raspberry Pi Ltd documentation. Image-specific checksum and filenames: output/image/FLASHING.md and SHA256SUMS-r4.